Lecture 5
Inner Product Spaces

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5.1 Length and Dot Product in R^n

- Length:

  The length of a vector $\mathbf{v} = (v_1, v_2, \cdots, v_n)$ in R^n is given by

$$\|\mathbf{v}\| = \sqrt{v_1^{\,2} + v_2^{\,2} + \cdots + v_n^{\,2}}$$

- Notes: The length of a vector is also called its **norm**.

- Notes: Properties of length

  (1) $\|\mathbf{v}\| \geq 0$

  (2) $\|\mathbf{v}\| = 1 \Rightarrow \mathbf{v}$ is called a **unit vector**.

  (3) $\|\mathbf{v}\| = 0$ iff $\mathbf{v} = (0,0,...,0)$

  (4) $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$

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- Ex 1:

(a) In R^3, the length of $\mathbf{v} = (\frac{2}{\sqrt{17}}, \frac{-2}{\sqrt{17}}, \frac{3}{\sqrt{17}})$ is given by

$$\|\mathbf{v}\| = \sqrt{\left(\frac{2}{\sqrt{17}}\right)^2 + \left(\frac{-2}{\sqrt{17}}\right)^2 + \left(\frac{3}{\sqrt{17}}\right)^2} = \sqrt{\frac{17}{17}} = 1$$

($\mathbf{v}$ is a unit vector)

(b) In R^5 the length of $\mathbf{v} = (0, -2, 1, 4, -2)$ is given by

$$\|\mathbf{v}\| = \sqrt{0^2 + (-2)^2 + 1^2 + 4^2 + (-2)^2} = \sqrt{25} = 5$$

- Standard unit vectors in R^n:

$$\{\mathbf{e}_1, \mathbf{e}_2, \cdots, \mathbf{e}_n\} = \{(1,0,\cdots,0), (0,1,\cdots,0), (0,0,\cdots,1)\}$$

- Ex:

the standard unit vector in R^2: $\{i, j\} = \{(1,0), (0,1)\}$

the standard unit vector in R^3: $\{i, j, k\} = \{(1,0,0), (0,1,0), (0,0,1)\}$

- Notes: (Two nonzero vectors are parallel)

$\mathbf{u} = c\mathbf{v}$

(1) $c > 0 \Rightarrow$ $\mathbf{u}$ and $\mathbf{v}$ have the same direction

(2) $c < 0 \Rightarrow$ $\mathbf{u}$ and $\mathbf{v}$ have the opposite direction

- Thm 5.1: (Length of a scalar multiple)

Let $\mathbf{v}$ be a vector in R^n and c be a scalar. Then

$$\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$$

Pf:

$$\mathbf{v} = (v_1, v_2, \cdots, v_n)$$
$$\Rightarrow c\mathbf{v} = (cv_1, cv_2, \cdots, cv_n)$$
$$\|c\mathbf{v}\| = \|(cv_1, cv_2, \cdots, cv_n)\|$$
$$= \sqrt{(cv_1)^2 + (cv_2)^2 + \cdots + (cv_n)^2}$$
$$= \sqrt{c^2(v_1^2 + v_2^2 + \cdots + v_n^2)}$$
$$= |c|\sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$
$$= |c|\|\mathbf{v}\|$$

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- Thm 5.2: (Unit vector in the direction of **v**)

If $\mathbf{v}$ is a nonzero vector in R^n, then the vector $\mathbf{u} = \dfrac{\mathbf{v}}{\|\mathbf{v}\|} = \dfrac{1}{\|\mathbf{v}\|}\mathbf{v}$ has length 1 and has the same direction as $\mathbf{v}$. This vector $\mathbf{u}$ is called the **unit vector in the direction of v**.

Pf:

$\mathbf{v}$ is nonzero $\Rightarrow \|\mathbf{v}\| \neq 0 \Rightarrow \dfrac{1}{\|\mathbf{v}\|} > 0$

$\Rightarrow \mathbf{u} = \dfrac{1}{\|\mathbf{v}\|}\mathbf{v}$ ($\mathbf{u}$ has the same direction as $\mathbf{v}$)

$$\|\mathbf{u}\| = \left\|\frac{\mathbf{v}}{\|\mathbf{v}\|}\right\| = \frac{1}{\|\mathbf{v}\|}\|\mathbf{v}\| = 1$$ (u has length 1)

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- Notes:

(1) The vector $\dfrac{\mathbf{v}}{\|\mathbf{v}\|}$ is called the unit vector in the direction of **v**.

(2) The process of finding the unit vector in the direction of **v** is called **normalizing** the vector **v.**

- Ex 2: (Finding a unit vector)

Find the unit vector in the direction of $\mathbf{v} = (3, -1, 2)$, and verify that this vector has length 1.

Sol:

$$\mathbf{v} = (3, -1, 2) \quad \Rightarrow \|\mathbf{v}\| = \sqrt{3^2 + (-1)^2 + 2^2} = \sqrt{14}$$

$$\Rightarrow \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{(3, -1, 2)}{\sqrt{3^2 + (-1)^2 + 2^2}} = \frac{1}{\sqrt{14}}(3, -1, 2) = \left(\frac{3}{\sqrt{14}}, \frac{-1}{\sqrt{14}}, \frac{2}{\sqrt{14}}\right)$$

$$\because \quad \sqrt{\left(\frac{3}{\sqrt{14}}\right)^2 + \left(\frac{-1}{\sqrt{14}}\right)^2 + \left(\frac{2}{\sqrt{14}}\right)^2} = \sqrt{\frac{14}{14}} = 1$$

$\therefore$ $\dfrac{\mathbf{v}}{\|\mathbf{v}\|}$ is a unit vector.

- Distance between two vectors:

  The **distance** between two vectors $\mathbf{u}$ and $\mathbf{v}$ in R^n is

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

- Notes: (Properties of distance)

  (1) $d(\mathbf{u}, \mathbf{v}) \geq 0$

  (2) $d(\mathbf{u}, \mathbf{v}) = 0$ if and only if $\mathbf{u} = \mathbf{v}$

  (3) $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$

- Ex 3: (Finding the distance between two vectors)

  The distance between $\mathbf{u}$=(0, 2, 2) and $\mathbf{v}$=(2, 0, 1) is

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \|(0-2, 2-0, 2-1)\|$$
$$= \sqrt{(-2)^2 + 2^2 + 1^2} = 3$$

- Dot product in R^n:

  The **dot product** of $\mathbf{u} = (u_1, u_2, \cdots, u_n)$ and $\mathbf{v} = (v_1, v_2, \cdots, v_n)$ is the scalar quantity

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

- Ex 4: (Finding the dot product of two vectors)

  The dot product of **u**=(1, 2, 0, -3) and **v**=(3, -2, 4, 2) is

$$\mathbf{u} \cdot \mathbf{v} = (1)(3) + (2)(-2) + (0)(4) + (-3)(2) = -7$$

- Thm 5.3: (Properties of the dot product)

  If **u**, **v**, and **w** are vectors in R^n and c is a scalar, then the following properties are true.

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$$

$$\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$$

$$c(\mathbf{u} \cdot \mathbf{v}) = (c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v})$$

$$\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$$

$$\mathbf{v} \cdot \mathbf{v} \geq 0, \text{ and } \mathbf{v} \cdot \mathbf{v} = 0 \text{ if and only if } \mathbf{v} = 0$$

- Euclidean *n*-space:

  R^n was defined to be the *set* of all order n-tuples of real numbers. When R^n is combined with the standard operations of vector addition, scalar multiplication, vector length, and the dot product, the resulting vector space is called **Euclidean *n*-space**.

- Ex 5: (Finding dot products)

$$\mathbf{u} = (2, -2),\ \mathbf{v} = (5, 8),\ \mathbf{w} = (-4, 3)$$

(a) $\mathbf{u} \cdot \mathbf{v}$ (b) $(\mathbf{u} \cdot \mathbf{v})\mathbf{w}$ (c) $\mathbf{u} \cdot (2\mathbf{v})$ (d) $\|\mathbf{w}\|^2$ (e) $\mathbf{u} \cdot (\mathbf{v} - 2\mathbf{w})$

Sol:

(a) $\mathbf{u} \cdot \mathbf{v} = (2)(5) + (-2)(8) = -6$

(b) $(\mathbf{u} \cdot \mathbf{v})\mathbf{w} = -6\mathbf{w} = -6(-4, 3) = (24, -18)$

(c) $\mathbf{u} \cdot (2\mathbf{v}) = 2(\mathbf{u} \cdot \mathbf{v}) = 2(-6) = -12$

(d) $\|\mathbf{w}\|^2 = \mathbf{w} \cdot \mathbf{w} = (-4)(-4) + (3)(3) = 25$

(e) $\mathbf{v} - 2\mathbf{w} = (5 - (-8), 8 - 6) = (13, 2)$

$\mathbf{u} \cdot (\mathbf{v} - 2\mathbf{w}) = (2)(13) + (-2)(2) = 26 - 4 = 22$

- Ex 6: (Using the properties of the dot product)

  Given $\mathbf{u}\cdot\mathbf{u}=39 \quad \mathbf{u}\cdot\mathbf{v}=-3 \quad \mathbf{v}\cdot\mathbf{v}=79$

  Find $(\mathbf{u}+2\mathbf{v})\cdot(3\mathbf{u}+\mathbf{v})$

Sol:

$$\begin{aligned}(\mathbf{u}+2\mathbf{v})\cdot(3\mathbf{u}+\mathbf{v}) &= \mathbf{u}\cdot(3\mathbf{u}+\mathbf{v})+2\mathbf{v}\cdot(3\mathbf{u}+\mathbf{v})\\ &= \mathbf{u}\cdot(3\mathbf{u})+\mathbf{u}\cdot\mathbf{v}+(2\mathbf{v})\cdot(3\mathbf{u})+(2\mathbf{v})\cdot\mathbf{v}\\ &= 3(\mathbf{u}\cdot\mathbf{u})+\mathbf{u}\cdot\mathbf{v}+6(\mathbf{v}\cdot\mathbf{u})+2(\mathbf{v}\cdot\mathbf{v})\\ &= 3(\mathbf{u}\cdot\mathbf{u})+7(\mathbf{u}\cdot\mathbf{v})+2(\mathbf{v}\cdot\mathbf{v})\\ &= 3(39)+7(-3)+2(79)=254\end{aligned}$$

- Thm 5.4: (The Cauchy - Schwarz inequality)

  If $\mathbf{u}$ and $\mathbf{v}$ are vectors in R^n, then

  $|\mathbf{u}\cdot\mathbf{v}|\le\|\mathbf{u}\|\|\mathbf{v}\|$ ($|\mathbf{u}\cdot\mathbf{v}|$ denotes the absolute value of $\mathbf{u}\cdot\mathbf{v}$)

- Ex 7: (An example of the Cauchy - Schwarz inequality)

  Verify the Cauchy - Schwarz inequality for $\mathbf{u}$=(1, -1, 3) and $\mathbf{v}$=(2, 0, -1)

Sol: $\mathbf{u}\cdot\mathbf{v}=-1, \quad \mathbf{u}\cdot\mathbf{u}=11, \quad \mathbf{v}\cdot\mathbf{v}=5$

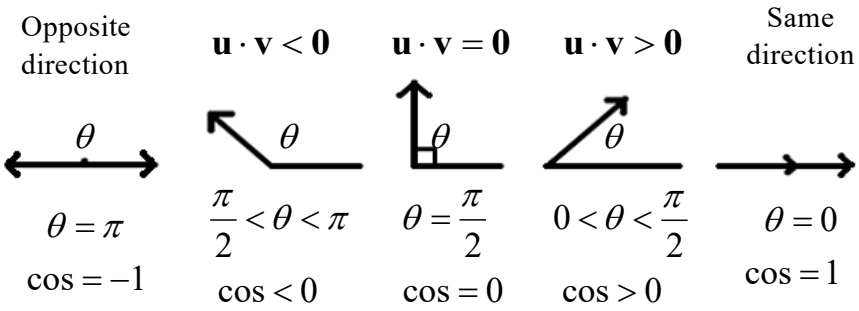
$$\Rightarrow |\mathbf{u}\cdot\mathbf{v}|=|-1|=1$$

$$\|\mathbf{u}\|\|\mathbf{v}\|=\sqrt{\mathbf{u}\cdot\mathbf{u}}\cdot\sqrt{\mathbf{v}\cdot\mathbf{v}}=\sqrt{11}\cdot\sqrt{5}=\sqrt{55}$$

$$\therefore |\mathbf{u}\cdot\mathbf{v}|\le\|\mathbf{u}\|\|\mathbf{v}\|$$

- The angle between two vectors **u** and **v** in R^n: is the angle θ satisfying

$$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|},\ 0 \le \theta \le \pi$$



- Note:

  The angle between the zero vector and another vector is not defined.

- Ex 8: (Finding the angle between two vectors)

  $\mathbf{u} = (-4, 0, 2, -2) \quad \mathbf{v} = (2, 0, -1, 1)$

Sol:

$$\|\mathbf{u}\| = \sqrt{\mathbf{u}\cdot\mathbf{u}} = \sqrt{(-4)^2 + 0^2 + 2^2 + (-2)^2} = \sqrt{24}$$

$$\|\mathbf{v}\| = \sqrt{\mathbf{v}\cdot\mathbf{v}} = \sqrt{2^2 + (0)^2 + (-1)^2 + 1^2} = \sqrt{6}$$

$$\mathbf{u}\cdot\mathbf{v} = (-4)(2) + (0)(0) + (2)(-1) + (-2)(1) = -12$$

$$\Rightarrow \cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} = \frac{-12}{\sqrt{24}\sqrt{6}} = -\frac{12}{\sqrt{144}} = -1$$

$\Rightarrow \theta = \pi \quad \therefore$ **u** and **v** have opposite directions. $(\mathbf{u} = -2\mathbf{v})$

- Orthogonal vectors:

  Two vectors **u** and **v** in R^n are orthogonal if

  $$\mathbf{u} \cdot \mathbf{v} = 0$$

- Note:

  The vector **0** is said to be orthogonal to every vector.

- Ex 10: (Finding orthogonal vectors)

  Determine all vectors in R^2 that are orthogonal to **u**=(4, 2).

Sol:

$$\mathbf{u} = (4, 2) \quad \text{Let} \quad \mathbf{v} = (v_1, v_2)$$

$$\Rightarrow \quad \mathbf{u} \cdot \mathbf{v} = (4, 2) \cdot (v_1, v_2)$$

$$= 4v_1 + 2v_2$$

$$= 0$$

$$\begin{bmatrix} 4 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{1}{2} & 0 \end{bmatrix}$$

$$\Rightarrow \quad v_1 = \frac{-t}{2} \quad , \quad v_2 = t$$

$$\therefore \quad \mathbf{v} = \left(\frac{-t}{2}, t \right), \qquad t \in R$$

- Thm 5.5: (The triangle inequality)

  If $\mathbf{u}$ and $\mathbf{v}$ are vectors in R^n, then $\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$

Pf:

$$\begin{aligned}\|\mathbf{u}+\mathbf{v}\|^2 &= (\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}+\mathbf{v}) \\ &= \mathbf{u}\cdot(\mathbf{u}+\mathbf{v}) + \mathbf{v}\cdot(\mathbf{u}+\mathbf{v}) = \mathbf{u}\cdot\mathbf{u} + 2(\mathbf{u}\cdot\mathbf{v}) + \mathbf{v}\cdot\mathbf{v} \\ &= \|\mathbf{u}\|^2 + 2(\mathbf{u}\cdot\mathbf{v}) + \|\mathbf{v}\|^2 \le \|\mathbf{u}\|^2 + 2|\mathbf{u}\cdot\mathbf{v}| + \|\mathbf{v}\|^2 \\ &\le \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 \\ &= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2\end{aligned}$$

$$\therefore \|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$$

- Note:

  Equality occurs in the triangle inequality if and only if the vectors $\mathbf{u}$ and $\mathbf{v}$ have the same direction.

- Thm 5.6: (The Pythagorean theorem)

  If $\mathbf{u}$ and $\mathbf{v}$ are vectors in R^n, then $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if

$$\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

- Dot product and matrix multiplication:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \qquad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

(A vector $\mathbf{u} = (u_1, u_2, \cdots, u_n)$ in R^n is represented as an $n \times 1$ column matrix)

$$[u_1 \quad u_2 \quad \cdots \quad u_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = [u_1 v_1 + u_2 v_2 + \cdots + u_n v_n]$$

$$\mathbf{u}^T \mathbf{v} \quad = \quad \mathbf{u} \cdot \mathbf{v}$$

5.2 Inner Product Spaces

- Inner product:

Let **u**, **v**, and **w** be vectors in a vector space V, and let c be any scalar. An inner product on V is a function that associates a real number <**u**, **v**> with each pair of vectors **u** and **v** and satisfies the following axioms.

(1) <**u**, **v**> = <**v**,**u**>

(2) <**u**, **v**+**w**> = <**u**, **v**> + <**u**, **w**>

(3) c <**u**, **v**> = < c **u**, **v**>

(4) <**v**,**v**> ≥ 0 and <**v**,**v**> = 0 if and only if **v** = 0

- Note:

$\mathbf{u} \cdot \mathbf{v}$ = dot product (Euclidean inner product for R^n)

$< \mathbf{u} , \mathbf{v} >$= general inner product for vector space V

- Note:

A vector space V with an inner product is called an inner product space.

Vector space: $(V, +, \bullet)$

Inner product space: $(V, +, \bullet, <,>)$

- Ex 1: (The Euclidean inner product for R^n)

Show that the dot product in R^n satisfies the four axioms of an inner product.

Sol:

$\mathbf{u} = (u_1 , u_2 , \cdots , u_n)$, $\mathbf{v} = (v_1 , v_2 , \cdots , v_n)$

$\langle \mathbf{u} , \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$

By Theorem 5.3, this dot product satisfies the required four axioms. Thus it is an inner product on R^n.

- Ex 2: (A different inner product for R^n)

Show that the function $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + 2u_2 v_2$ defines an inner product on R^2, where $\mathbf{u} = (u_1, u_2)$ $\mathbf{v} = (v_1, v_2)$

Remark: *This is not the usual dot product in R^2*

Sol:

(a) $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + 2u_2 v_2 = v_1 u_1 + 2v_2 u_2 = \langle \mathbf{v}, \mathbf{u} \rangle$

(b) $\mathbf{w} = (w_1, w_2)$

$$\Rightarrow \langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = u_1(v_1 + w_1) + 2u_2(v_2 + w_2)$$
$$= u_1 v_1 + u_1 w_1 + 2u_2 v_2 + 2u_2 w_2$$
$$= (u_1 v_1 + 2u_2 v_2) + (u_1 w_1 + 2u_2 w_2)$$
$$= \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$$

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(c) $c\langle \mathbf{u}, \mathbf{v} \rangle = c(u_1 v_1 + 2u_2 v_2) = (cu_1)v_1 + 2(cu_2)v_2 = \langle c\mathbf{u}, \mathbf{v} \rangle$

(d) $\langle \mathbf{v}, \mathbf{v} \rangle = v_1^2 + 2v_2^2 \geq 0$

$\langle \mathbf{v}, \mathbf{v} \rangle = 0 \Rightarrow v_1^2 + 2v_2^2 = 0 \quad \Rightarrow \quad v_1 = v_2 = 0 \quad (\mathbf{v} = 0)$

- Note: (An inner product on R^n)

$$\langle \mathbf{u}, \mathbf{v} \rangle = c_1 u_1 v_1 + c_2 u_2 v_2 + \cdots + c_n u_n v_n, \qquad c_i > 0$$

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- Ex 3: (A function that is not an inner product)

Show that the following function is not an inner product on R^3.

$$\langle \mathbf{u} \cdot \mathbf{v} \rangle = u_1 v_1 - 2u_2 v_2 + u_3 v_3$$

Sol:

Let $\mathbf{v} = (1, 2, 1)$

Then $< \mathbf{v}, \mathbf{v} >= (1)(1) - 2(2)(2) + (1)(1) = -6 < 0$

Axiom 4 is not satisfied.

Thus this function is not an inner product on R^3.

- Thm 5.7: (Properties of inner products)

Let **u, v**, and **w** be vectors in an inner product space V, and let c be any real number. Then

(1) $<\mathbf{0}, \mathbf{v}> = <\mathbf{v}, \mathbf{0}> = 0$

(2) $<\mathbf{u} + \mathbf{v}, \mathbf{w}> = <\mathbf{u}, \mathbf{w}> + <\mathbf{v}, \mathbf{w}>$

(3) $<\mathbf{u}, c\mathbf{v}> = c <\mathbf{u}, \mathbf{v}>$

- Norm (length) of **u**:

$$\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$$

- Note:

$$\|\mathbf{u}\|^2 = \langle \mathbf{u}, \mathbf{u} \rangle$$

- Distance between **u** and **v**:

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{\langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle}$$

- Angle between two nonzero vectors **u** and **v**:

$$\cos\theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}, \quad 0 \le \theta \le \pi$$

- Orthogonal: $(\mathbf{u} \perp \mathbf{v})$

u and **v** are **orthogonal** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

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- Notes:

(1) If $\|\mathbf{v}\| = 1$, then v is called a unit vector.

(2) $\|\mathbf{v}\| \neq 1$, $\mathbf{v} \neq 0$ $\xrightarrow{\text{Normalizing}}$ $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ (the unit vector in the direction of **v**)

not a unit vector

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- Ex 6: *Given* $p(x)=a_0+a_1x+a_2x^2+\ldots+a_nx^n$, $q(x)=b_0+b_1x+b_2x^2+\ldots+b_nx^n$.

Then $\langle p\,,\,q\rangle = a_0b_0 + a_1b_1 + \cdots + a_nb_n$ is an inner product in $P_n(x)$

Let $p(x)=1-2x^2$, $q(x)=4-2x+x^2$ be polynomials in $P_2(x)$

(a) $\langle p\,,\,q\rangle = ?$ (b) $\|q\| = ?$ (c) $d(p\,,\,q) = ?$

Sol:

(a) $\langle p\,,\,q\rangle = (1)(4)+(0)(-2)+(-2)(1) = 2$

(b) $\|q\| = \sqrt{\langle q\,,\,q\rangle} = \sqrt{4^2+(-2)^2+1^2} = \sqrt{21}$

(c) $\because\ p-q = -3+2x-3x^2$

$\therefore d(p\,,\,q) = \|p-q\| = \sqrt{\langle p-q, p-q\rangle}$

$= \sqrt{(-3)^2+2^2+(-3)^2} = \sqrt{22}$

- Properties of norm:

(1) $\|\mathbf{u}\| \geq 0$

(2) $\|\mathbf{u}\| = 0$ if and only if $\mathbf{u} = \mathbf{0}$

(3) $\|c\mathbf{u}\| = |c|\|\mathbf{u}\|$

- Properties of distance:

(1) $d(\mathbf{u}\,,\,\mathbf{v}) \geq 0$

(2) $d(\mathbf{u}\,,\,\mathbf{v}) = 0$ if and only if $\mathbf{u} = \mathbf{v}$

(3) $d(\mathbf{u}\,,\,\mathbf{v}) = d(\mathbf{v}\,,\,\mathbf{u})$

- Thm 5.8：Summary

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space V.

(1) Cauchy-Schwarz inequality:

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \le \|\mathbf{u}\| \|\mathbf{v}\| \qquad \text{Theorem 5.4}$$

(2) Triangle inequality:

$$\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\| \qquad \text{Theorem 5.5}$$

(3) Pythagorean theorem :

$\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if

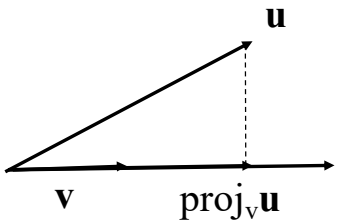
$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \qquad \text{Theorem 5.6}$$

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- Orthogonal projections in inner product spaces:

Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors in an inner product space V, such that $\mathbf{v} \neq \mathbf{0}$. Then the **orthogonal projection of u onto v** is a vector in V, given by

$$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle}\mathbf{v}$$



- Note:

If $\mathbf{v}$ is a unit vector, then $\langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{v}\|^2 = 1$

The formula for the orthogonal projection of $\mathbf{u}$ onto $\mathbf{v}$ takes the following simpler form.

$$\text{proj}_{\mathbf{v}}\mathbf{u} = \langle \mathbf{u}, \mathbf{v} \rangle \mathbf{v}$$

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- Ex 10: (Finding an orthogonal projection in R^3)

  Use the Euclidean inner product in R^3 to find the orthogonal projection of $\mathbf{u}=(6, 2, 4)$ onto $\mathbf{v}=(1, 2, 0)$.

Sol:

$$\because \langle \mathbf{u}, \mathbf{v} \rangle = (6)(1)+(2)(2)+(4)(0) = 10$$

$$\langle \mathbf{v}, \mathbf{v} \rangle = 1^2 + 2^2 + 0^2 = 5$$

$$\therefore \text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v} = \tfrac{10}{5}(1, 2, 0) = (2, 4, 0)$$

- Note:

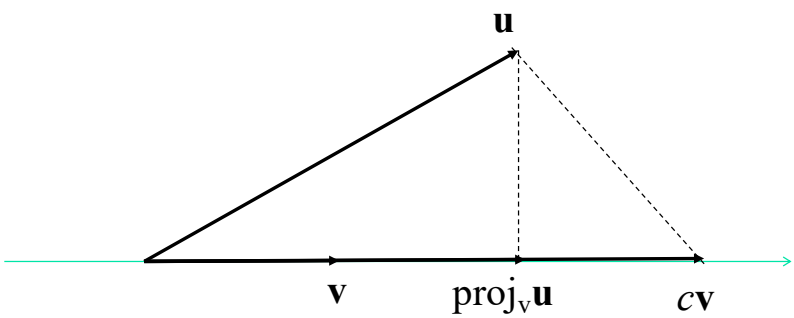
$\mathbf{u} - \text{proj}_{\mathbf{v}}\mathbf{u} = (6, 2, 4) - (2, 4, 0) = (4, -2, 4)$ is orthogonal to $\mathbf{v} = (1,2,0)$.

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- Thm 5.9: (Orthogonal projection and distance)

  Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors in an inner product space V, such that $\mathbf{v} \neq \mathbf{0}$. Then

$$d(\mathbf{u}, \text{proj}_{\mathbf{v}}\mathbf{u}) < d(\mathbf{u}, c\mathbf{v}), \qquad c \neq \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle}$$



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5.3 Orthonormal Bases: Gram-Schmidt Process

- Orthogonal:

  A set S of vectors in an inner product space V is called an **orthogonal set** if every pair of vectors in the set is orthogonal.

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\} \subseteq V$$
$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0 \quad i \neq j$$

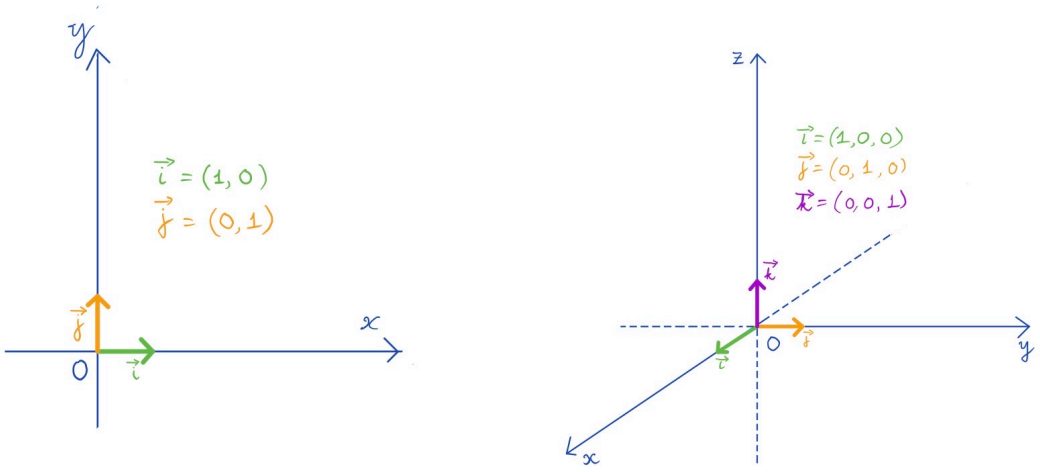
- Orthonormal:

  An orthogonal set in which each vector is a unit vector is called **orthonormal**.

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\} \subseteq V$$
$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

- Note:

  If S is a basis, then it is called an orthogonal basis or an orthonormal basis.



- Ex 1: (A nonstandard orthonormal basis for R^3)

Show that the following set is an orthonormal basis.

$$S = \{\underbrace{\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)}_{\mathbf{v}_1}, \underbrace{\left(-\frac{\sqrt{2}}{6}, \frac{\sqrt{2}}{6}, \frac{2\sqrt{2}}{3}\right)}_{\mathbf{v}_2}, \underbrace{\left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}\right)}_{\mathbf{v}_3}\}$$

Sol:

We show that three vectors are mutually orthogonal:

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = -\tfrac{1}{6} + \tfrac{1}{6} + 0 = 0$$

$$\mathbf{v}_1 \cdot \mathbf{v}_3 = \frac{2}{3\sqrt{2}} - \frac{2}{3\sqrt{2}} + 0 = 0$$

$$\mathbf{v}_2 \cdot \mathbf{v}_3 = -\frac{\sqrt{2}}{9} - \frac{\sqrt{2}}{9} + \frac{2\sqrt{2}}{9} = 0$$

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We show that each vector is of length 1.

$$\|\mathbf{v}_1\| = \sqrt{\mathbf{v}_1 \cdot \mathbf{v}_1} = \sqrt{\tfrac{1}{2} + \tfrac{1}{2} + 0} = 1$$

$$\|\mathbf{v}_2\| = \sqrt{\mathbf{v}_2 \cdot \mathbf{v}_2} = \sqrt{\tfrac{2}{36} + \tfrac{2}{36} + \tfrac{8}{9}} = 1$$

$$\|\mathbf{v}_3\| = \sqrt{\mathbf{v}_3 \cdot \mathbf{v}_3} = \sqrt{\tfrac{4}{9} + \tfrac{4}{9} + \tfrac{1}{9}} = 1$$

Thus S is an orthonormal set.

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- Ex 2: (An orthonormal basis for $P_3(x)$)

In $P_3(x)$, with the inner product

$\langle p, q\rangle = a_0b_0 + a_1b_1 + a_2b_2$ for $p=a_0+a_1x+a_2x^2$ and $q = b_0+b_1x+b_2x^2$

The standard basis $B = \{1, x, x^2\}$ is orthonormal.

Sol:

$$\mathbf{v}_1 = 1 + 0x + 0x^2, \quad \mathbf{v}_2 = 0 + x + 0x^2, \quad \mathbf{v}_3 = 0 + 0x + x^2,$$

Then

$$\langle \mathbf{v}_1, \mathbf{v}_2\rangle = (1)(0) + (0)(1) + (0)(0) = 0,$$
$$\langle \mathbf{v}_1, \mathbf{v}_3\rangle = (1)(0) + (0)(0) + (0)(1) = 0,$$
$$\langle \mathbf{v}_2, \mathbf{v}_3\rangle = (0)(0) + (1)(0) + (0)(1) = 0$$

$$\|\mathbf{v}_1\| = \sqrt{\langle \mathbf{v}_1, \mathbf{v}_1\rangle} = \sqrt{(1)(1)+(0)(0)+(0)(0)} = 1,$$
$$\|\mathbf{v}_2\| = \sqrt{\langle \mathbf{v}_2, \mathbf{v}_2\rangle} = \sqrt{(0)(0)+(1)(1)+(0)(0)} = 1,$$
$$\|\mathbf{v}_3\| = \sqrt{\langle \mathbf{v}_3, \mathbf{v}_3\rangle} = \sqrt{(0)(0)+(0)(0)+(1)(1)} = 1$$

Homework #2: Show that $B = \{1, x, x^2, \ldots, x^{n-1}, x^n\}$ is an orthonormal basis for $P_n(x)$.

- Thm 5.10: (Orthogonal sets are linearly independent)

  If $S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is an orthogonal set of nonzero vectors in an inner product space V, then S is linearly independent.

Pf:

S is an orthogonal set of nonzero vectors

i.e. $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0 \quad i \neq j$ and $\langle \mathbf{v}_i, \mathbf{v}_i \rangle > 0$

Let $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = 0$

$\Rightarrow \quad \langle c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n, \mathbf{v}_i \rangle = \langle 0, \mathbf{v}_i \rangle = 0 \quad \forall i$

$\Rightarrow \quad c_1\langle \mathbf{v}_1, \mathbf{v}_i \rangle + c_2\langle \mathbf{v}_2, \mathbf{v}_i \rangle + \cdots + c_i\langle \mathbf{v}_i, \mathbf{v}_i \rangle + \cdots + c_n\langle \mathbf{v}_n, \mathbf{v}_i \rangle$

$= c_i\langle \mathbf{v}_i, \mathbf{v}_i \rangle$

$\because \langle \mathbf{v}_i, \mathbf{v}_i \rangle \neq 0 \Rightarrow c_i = 0 \quad \forall i \quad \therefore S$ is linearly independent.

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- Corollary to Thm 5.10:

  If V is an inner product space of dimension n, then any orthogonal set of n nonzero vectors is a basis for V.

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- Ex 4: (Using orthogonality to test for a basis)

Show that the following set is a basis for R^4.

$$S = \{\underbrace{(2,3,2,-2)}_{\mathbf{v}_1}, \underbrace{(1,0,0,1)}_{\mathbf{v}_2}, \underbrace{(-1,0,2,1)}_{\mathbf{v}_3}, \underbrace{(-1,2,-1,1)}_{\mathbf{v}_4}\}$$

Sol:

$\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$: nonzero vectors

$\mathbf{v}_1 \cdot \mathbf{v}_2 = 2+0+0-2 = 0$ $\quad$ $\mathbf{v}_2 \cdot \mathbf{v}_3 = -1+0+0+1 = 0$

$\mathbf{v}_1 \cdot \mathbf{v}_3 = -2+0+4-2 = 0$ $\quad$ $\mathbf{v}_2 \cdot \mathbf{v}_4 = -1+0+0+1 = 0$

$\mathbf{v}_1 \cdot \mathbf{v}_4 = -2+6-2-2 = 0$ $\quad$ $\mathbf{v}_3 \cdot \mathbf{v}_4 = 1+0-2+1 = 0$

$\Rightarrow S$ is orthogonal.

$\Rightarrow S$ is a basis for R^4 (by Corollary to Theorem 5.10)

- Thm 5.11: (Coordinates relative to an orthonormal basis)

If $B = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is an orthonormal basis for an inner product space V, then the coordinate representation of a vector $\mathbf{w}$ with respect to B is

$$\mathbf{w} = \langle \mathbf{w}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{w}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \cdots + \langle \mathbf{w}, \mathbf{v}_n \rangle \mathbf{v}_n$$

Pf:

$B = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is a basis for V

$\mathbf{w} \in V$

$\mathbf{w} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \cdots + k_n\mathbf{v}_n$ (unique representation)

$\because B = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is orthonormal

$$\Rightarrow \langle \mathbf{v}_i, \mathbf{v}_j \rangle = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

$$\langle \mathbf{w},\mathbf{v}_i \rangle = \langle (k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \ldots + k_n\mathbf{v}_n), \mathbf{v}_i \rangle$$
$$= k_1 \langle \mathbf{v}_1 , \mathbf{v}_i \rangle + \ldots + k_i \langle \mathbf{v}_i , \mathbf{v}_i \rangle + \ldots + k_n \langle \mathbf{v}_n , \mathbf{v}_i \rangle$$
$$= k_i \text{ for all i.}$$
$$\Rightarrow \mathbf{w} = \langle \mathbf{w}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{w}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \cdots + \langle \mathbf{w}, \mathbf{v}_n \rangle \mathbf{v}_n$$

- Note:

If $B = \{\mathbf{v}_1 , \mathbf{v}_2 , \cdots , \mathbf{v}_n\}$ is an orthonormal basis for V and $\mathbf{w} \in V$,

Then the corresponding coordinate matrix of $\mathbf{w}$ relative to B is

$$[\mathbf{w}]_B = \begin{bmatrix} \langle \mathbf{w}, \mathbf{v}_1 \rangle \\ \langle \mathbf{w}, \mathbf{v}_2 \rangle \\ \vdots \\ \langle \mathbf{w}, \mathbf{v}_n \rangle \end{bmatrix}$$

- Ex 5: (Representing vectors relative to an orthonormal basis)

Find the coordinates of $\mathbf{w} = (5, -5, 2)$ relative to the following orthonormal basis for R^3.

$$B = \{(\tfrac{3}{5} , \tfrac{4}{5} , 0) , (-\tfrac{4}{5} , \tfrac{3}{5} , 0) , (0 , 0 , 1)\}$$

Sol:

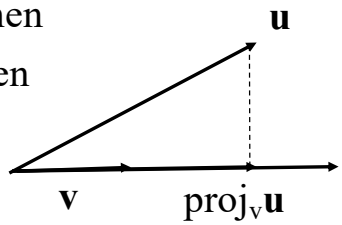
$$\langle \mathbf{w}, \mathbf{v}_1 \rangle = \mathbf{w} \cdot \mathbf{v}_1 = (5 , -5 , 2) \cdot (\tfrac{3}{5} , \tfrac{4}{5} , 0) = -1$$
$$\langle \mathbf{w}, \mathbf{v}_2 \rangle = \mathbf{w} \cdot \mathbf{v}_2 = (5, -5 , 2) \cdot (-\tfrac{4}{5} , \tfrac{3}{5} , 0) = -7$$
$$\langle \mathbf{w}, \mathbf{v}_3 \rangle = \mathbf{w} \cdot \mathbf{v}_3 = (5 , -5 , 2) \cdot (0 , 0 , 1) = 2$$

$$\Rightarrow [\mathbf{w}]_B = \begin{bmatrix} -1 \\ -7 \\ 2 \end{bmatrix}$$

- Orthogonal projections in inner product spaces:

Asume that u, v are vectors in the space V. Then the **orthogonal projection of u onto v** is given by

$$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle}\mathbf{v}$$

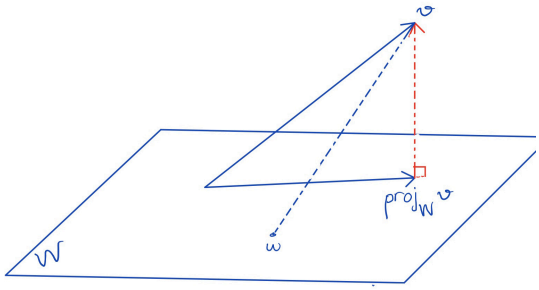


Property: $\mathbf{u}$ - $\text{proj}_{\mathbf{v}}\mathbf{u}$ is orthogonal to $\mathbf{v}$

Assume that W is a subspace of V. The **orthogonal projection of v onto W** is the vector denoted by $\text{proj}_{W}\mathbf{v}$ in W such that

$\mathbf{v}$-$\text{proj}_{W}\mathbf{v}$ is orthogonal to W

$v - \text{proj}_W v \perp W$



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- Gram-Schmidt orthonormalization process:

$B = \{\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_n\}$ is a basis for an inner product space V

Let $\mathbf{v}_1 = \mathbf{u}_1$ $\quad W_1 = span(\{\mathbf{v}_1\})$

$$\mathbf{v}_2 = \mathbf{u}_2 - \text{proj}_{W_1}\mathbf{u}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle}\mathbf{v}_1 \qquad W_2 = span(\{\mathbf{v}_1, \mathbf{v}_2\})$$

$$\mathbf{v}_3 = \mathbf{u}_3 - \text{proj}_{W_2}\mathbf{u}_3 = \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle}\mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\langle \mathbf{v}_2, \mathbf{v}_2 \rangle}\mathbf{v}_2$$

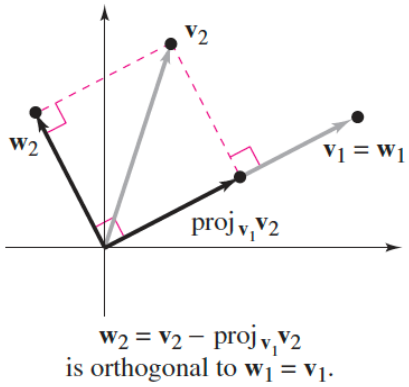
$$\vdots$$

$$\mathbf{v}_n = \mathbf{u}_n - \text{proj}_{W_{n-1}}\mathbf{u}_n = \mathbf{u}_n - \sum_{i=1}^{n-1} \frac{\langle \mathbf{v}_n, \mathbf{v}_i \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle}\mathbf{v}_i$$

$\Rightarrow B' = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is an orthogonal basis.

$\Rightarrow B'' = \{\frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}, \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|}, \cdots, \frac{\mathbf{v}_n}{\|\mathbf{v}_n\|}\}$ is an orthonormal basis.

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$w_2 = v_2 - \text{proj}_{v_1} v_2$ is orthogonal to $w_1 = v_1$.

Let $B = \{v_1, v_2\}$ is a basis of R^2. This is not yet an orthonormal basis.

Applying the Gram-Schmidt process to find an orthonormal basis for R^2.

1st step: take $w_1 = v_1$

2nd step: choose $w_2 = v_2 - proj_{v_1} v_2$

Obviously, $w_2 \perp w_1$.

Conclusion: $\{u_1, u_2\} = \left\{\frac{w_1}{\|w_1\|}, \frac{w_2}{\|w_2\|}\right\}$ is an orthonormal basis of R^2

- Ex 7: (Applying the Gram-Schmidt orthonormalization process)

Apply the Gram-Schmidt process to the following basis.

$$B = \{\underbrace{(1,1,0)}_{\mathbf{u}_1}, \underbrace{(1,2,0)}_{\mathbf{u}_2}, \underbrace{(0,1,2)}_{\mathbf{u}_3}\}$$

Sol: $\mathbf{v}_1 = \mathbf{u}_1 = (1,1,0)$

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 = (1,2,0) - \frac{3}{2}(1,1,0) = (-\frac{1}{2}, \frac{1}{2}, 0)$$

$$\mathbf{v}_3 = \mathbf{u}_3 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2$$

$$= (0,1,2) - \frac{1}{2}(1,1,0) - \frac{1/2}{1/2}(-\frac{1}{2}, \frac{1}{2}, 0) = (0,0,2)$$

Orthogonal basis

$$\Rightarrow B' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{(1,1,0), (\frac{-1}{2}, \frac{1}{2}, 0), (0,0,2)\}$$

Orthonormal basis

$$\Rightarrow B'' = \{\frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}, \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|}, \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|}\} = \{(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0), (\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0), (0,0,1)\}$$

- Ex 10: (Alternative form of Gram-Schmidt orthonormalization process)

Find an orthonormal basis for the solution space of the homogeneous system of linear equations.

$$\begin{aligned} x_1 + x_2 \qquad\quad + 7x_4 &= 0 \\ 2x_1 + x_2 + 2x_3 + 6x_4 &= 0 \end{aligned}$$

Sol:

$$\begin{bmatrix} 1 & 1 & 0 & 7 & 0 \\ 2 & 1 & 2 & 6 & 0 \end{bmatrix} \xrightarrow{G.J.E} \begin{bmatrix} 1 & 0 & 2 & -1 & 0 \\ 0 & 1 & -2 & 8 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2s+t \\ 2s-8t \\ s \\ t \end{bmatrix} = s\begin{bmatrix} -2 \\ 2 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} 1 \\ -8 \\ 0 \\ 1 \end{bmatrix}$$

Thus one basis for the solution space is

$$B = \{\boldsymbol{u}_1, \boldsymbol{u}_2\} = \{(-2, 2, 1, 0), (1, -8, 0, 1)\}$$

$$\boldsymbol{v}_1 = \boldsymbol{u}_1 = (-2, 2, 1, 0)$$

$$\boldsymbol{v}_2 = \boldsymbol{u}_2 - \frac{\langle \boldsymbol{u}_2, \boldsymbol{v}_1 \rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle} \boldsymbol{v}_1 = (1, -8, 0, 1) - \frac{-18}{9}(-2, 2, 1, 0)$$

$$= (-3, -4, 2, 1)$$

$$\Rightarrow B' = \{(-2,2,1,0)(-3,-4,2,1)\}$$ (orthogonal basis)

$$\Rightarrow B'' = \left\{\left(\frac{-2}{3}, \frac{2}{3}, \frac{1}{3}, 0\right), \left(\frac{-3}{\sqrt{30}}, \frac{-4}{\sqrt{30}}, \frac{2}{\sqrt{30}}, \frac{1}{\sqrt{30}}\right)\right\}$$

(orthonormal basis)

5.4 Mathematical Models and Least Squares Analysis

- Orthogonal subspaces:

The subspaces W_1 and W_2 of an inner product space V are orthogonal if $\langle \boldsymbol{v}_1, \boldsymbol{v}_2 \rangle = 0$ for all $\boldsymbol{v}_1$ in W_1 and all $\boldsymbol{v}_2$ in W_2.

- Ex 2: (Orthogonal subspaces) – Homework #3

The subspaces

$$W_1 = \text{span}\left(\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}\right) \text{ and } W_2 = \text{span}\left(\begin{bmatrix}-1\\1\\1\end{bmatrix}\right)$$

are orthogonal because $\langle \boldsymbol{v}_1, \boldsymbol{v}_2 \rangle = 0$ for any vector in W_1 and any vector in W_2 is zero.

- Orthogonal complement of W:

Let W be a subspace of an inner product space V.

(a) A vector $\mathbf{u}$ in V is said to **orthogonal to W**, if $\mathbf{u}$ is orthogonal to every vector in W.

(b) The set of all vectors in V that are orthogonal to every vector in W is called the **orthogonal complement of W**.

$$W^{\perp} = \{\mathbf{v} \in V \mid \langle \mathbf{v}, \mathbf{w} \rangle = 0, \forall\, \mathbf{w} \in W\}$$

$W^{\perp}$ (read "W perp")

- Notes:

(1) $(\{0\})^{\perp} = V$ (2) $V^{\perp} = \{0\}$

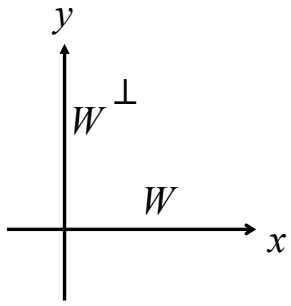
- Notes:

Suppose that W is a subspace of V

(1) $W^{\perp}$ is a subspace of V

(2) $W \cap W^{\perp} = \{\mathbf{0}\}$

(3) $(W^{\perp})^{\perp} = W$



Question: $W^{\perp} = ?$

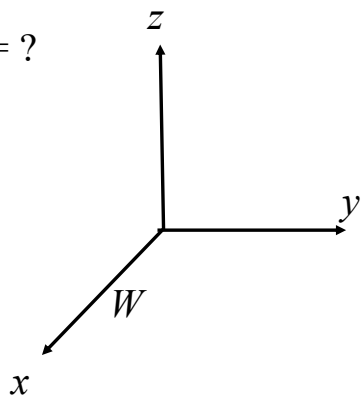
- Ex:

If $V = R^2$, $W = x-$axis

Then (1) $W^{\perp} = y$-axis is a subspace of R^2

(2) $W \cap W^{\perp} = \{(0,0)\}$

(3) $(W^{\perp})^{\perp} = W$



- Direct sum:

  Let W_1 and W_2 be two subspaces of R^n.

  If each vector $\mathbf{x} \in R^n$ can be **uniquely** written as a sum of a vector $\mathbf{w}_1$ from W_1 and a vector $\mathbf{w}_2$ from W_2, $\mathbf{x} = \mathbf{w}_1 + \mathbf{w}_2$, then R^n is called the direct sum of W_1 and W_2, and we write

$$R^n = W_1 \oplus W_2$$

- Thm 5.13: (Properties of orthogonal subspaces)

  Let W be a subspace of R^n. Then the following properties are true.

  (1) $\dim(W) + \dim(W^\perp) = n$

  (2) $R^n = W \oplus W^\perp$

  (3) $(W^\perp)^\perp = W$

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- Thm 5.14: (Projection onto a subspace)

  If $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_t\}$ is an orthonormal basis for the subspace W of V, and $\mathbf{v} \in V$, then

$$\text{proj}_W \mathbf{v} = \langle \mathbf{v}, \mathbf{u}_1 \rangle \mathbf{u}_1 + \langle \mathbf{v}, \mathbf{u}_2 \rangle \mathbf{u}_2 + \cdots + \langle \mathbf{v}, \mathbf{u}_t \rangle \mathbf{u}_t$$

Pf:

$\because \text{proj}_W \boldsymbol{v} \in \text{W}$ and $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_t\}$ is an orthonormal basis for W

$$\Rightarrow \text{proj}_W \boldsymbol{v} = \langle \text{proj}_W \boldsymbol{v}, \boldsymbol{u}_1 \rangle \boldsymbol{u}_1 + \cdots + \langle \text{proj}_W \boldsymbol{v}, \boldsymbol{u}_t \rangle \boldsymbol{u}_t$$

$$= \langle \boldsymbol{v} - \text{proj}_{W^\perp} \boldsymbol{v}, \boldsymbol{u}_1 \rangle \boldsymbol{u}_1 + \cdots + \langle \boldsymbol{v} - \text{proj}_{W^\perp} \boldsymbol{v}, \boldsymbol{u}_t \rangle \boldsymbol{u}_t$$

$$(\because \text{proj}_W \boldsymbol{v} = \boldsymbol{v} - \text{proj}_{W^\perp} \boldsymbol{v})$$

$$= \langle \boldsymbol{v}, \boldsymbol{u}_1 \rangle \boldsymbol{u}_1 + \cdots + \langle \boldsymbol{v}, \boldsymbol{u}_t \rangle \boldsymbol{u}_t \quad (\because \langle \text{proj}_{W^\perp} \boldsymbol{v}, \boldsymbol{u}_i \rangle = 0, \forall\, i)$$

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- Ex 5: (Projection onto a subspace)

$\mathbf{w}_1 = (0, 3, 1), \mathbf{w}_2 = (2, 0, 0), \mathbf{v} = (1, 1, 3)$

Find the projection of the vector $\mathbf{v}$ onto the subspace W

Sol: $W = span(\{\mathbf{w}_1, \mathbf{w}_2\})$

$\{\mathbf{w}_1, \mathbf{w}_2\}$: an orthogonal basis for W

$$\Rightarrow \{\mathbf{u}_1, \mathbf{u}_2\} = \left\{\frac{\mathbf{w}_1}{\|\mathbf{w}_1\|}, \frac{\mathbf{w}_2}{\|\mathbf{w}_2\|}\right\} = \left\{(0, \frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}}), (1,0,0)\right\} :$$

an orthonormal basis for W

$$\text{proj}_W \boldsymbol{v} = \langle \boldsymbol{v}, \boldsymbol{u}_1 \rangle \boldsymbol{u}_1 + \langle \boldsymbol{v}, \boldsymbol{u}_2 \rangle \boldsymbol{u}_2$$

$$= \frac{6}{\sqrt{10}}(0, \frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}}) + (1,0,0) = (1, \frac{9}{5}, \frac{3}{5})$$

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- Thm 5.15: (Orthogonal projection and distance)

Let W be a subspace of an inner product space V, and $\mathbf{v} \in V$.

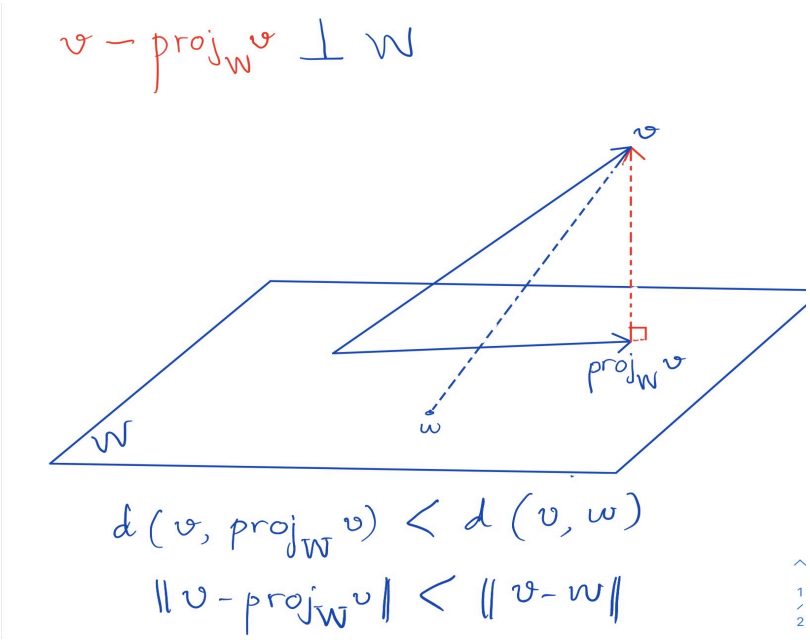
Then for all $\mathbf{w} \in W$, $\mathbf{w} \neq \text{proj}_W \mathbf{v}$

$$\|\mathbf{v} - \text{proj}_W \mathbf{v}\| < \|\mathbf{v} - \mathbf{w}\|$$

or $\|\boldsymbol{v} - \text{proj}_W \boldsymbol{v}\| = \min_{\boldsymbol{w} \in W} \|\boldsymbol{v} - \boldsymbol{w}\|$

($\text{proj}_W \mathbf{v}$ is the best approximation to $\mathbf{v}$ from W)

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- Pf:

$$\mathbf{v}-\mathbf{w}=(\mathbf{v}-\text{proj}_W\mathbf{v})+(\text{proj}_W\mathbf{v}-\mathbf{w})$$

$$(\mathbf{v}-\text{proj}_W\mathbf{v})\perp(\text{proj}_W\mathbf{v}-\mathbf{w})$$

By the Pythagorean theorem

$$\Rightarrow\|\mathbf{v}-\mathbf{w}\|^2=\|\mathbf{v}-\text{proj}_W\mathbf{v}\|^2+\|\text{proj}_W\mathbf{v}-\mathbf{w}\|^2$$

$$\mathbf{w}\neq\text{proj}_W\mathbf{v}\Rightarrow\|\text{proj}_W\mathbf{v}-\mathbf{w}\|>0$$

$$\Rightarrow\|\mathbf{v}-\mathbf{w}\|^2>\|\mathbf{v}-\text{proj}_W\mathbf{v}\|^2$$

$$\Rightarrow\|\mathbf{v}-\text{proj}_W\mathbf{v}\|<\|\mathbf{v}-\mathbf{w}\|$$

- Notes:

  (1) Among all the scalar multiples of a vector **u**, the orthogonal projection of **v** onto **u** is the one that is closest to **v**.

  (2) Among all the vectors in the subspace W, the vector $\text{proj}_W \mathbf{v}$ is the closest vector to **v**.

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- Thm 5.16: (Fundamental subspaces of a matrix)

  If A is an $m \times n$ matrix, then

  (1) $(CS(A))^{\perp} = NS(A^{\mathrm{T}})$ (2) $(CS(A^{\mathrm{T}}))^{\perp} = NS(A)$

  $(NS(A^{\mathrm{T}}))^{\perp} = CS(A)$ $(NS(A))^{\perp} = CS(A^{\mathrm{T}})$

  (3) $CS(A) \oplus NS(A^{T}) = R^{m}$ $CS(A) \oplus (NS(A))^{\perp} = R^{m}$

  (4) $CS(A^{T}) \oplus NS(A) = R^{n}$ $CS(A^{T}) \oplus (CS(A^{\mathrm{T}}))^{\perp} = R^{n}$

Recall: If A is an $m \times n$ matrix, then

$\mathrm{CS(A)} \subseteq \mathrm{R^m}$ $\mathrm{CS(A^T)} = \mathrm{RS(A)} \subseteq \mathrm{R^n}$

$\mathrm{NS(A^T)} \subseteq \mathrm{R^m}$ $\mathrm{NS(A)} \subseteq \mathrm{R^n}$

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- Ex 6: (Fundamental subspaces)

Find the four fundamental subspaces of the matrix.

$$A=\begin{bmatrix}1 & 2 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix} \quad \text{(reduced row-echelon form)}$$

Sol:

$$CS(A)=\text{span}(\{(1,0,0,0)\ \ (0,1,0,0)\}) \ \text{ is a subspace of } R^4$$

$$CS(A^{\mathrm{T}})=RS(A)=\text{span}(\{(1,2,0)\ \ (0,0,1)\}) \ \text{ is a subspace of } R^3$$

$$NS(A)=\text{span}(\{(-2,1,0)\}) \ \text{ is a subspace of } R^3$$

$$A^{\mathrm{T}}=\begin{bmatrix}1 & 0 & 0 & 0\\ 2 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\end{bmatrix}\sim R=\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0\\ & & s & t\end{bmatrix}$$

$$NS(A^{\mathrm{T}})=\text{span}(\{(0,0,1,0)\ \ (0,0,0,1)\}) \ \text{ is a subspace of } R^4$$

- Check:

$$(CS(A))^{\perp}=NS(A^{\mathrm{T}})$$

$$(CS(A^{\mathrm{T}}))^{\perp}=NS(A)$$

$$CS(A)\oplus NS(A^{T})=R^4$$

$$CS(A^{T})\oplus NS(A)=R^3$$

- Ex 3 & Ex 4:

Let $W = span(\{\mathbf{w}_1, \mathbf{w}_2\})$ is a subspace of R^4 , where

$\mathbf{w}_1 = (1, 2, 1, 0), \quad \mathbf{w}_2 = (0, 0, 0, 1)$

(a) Find a basis for W

(b) Find a basis for the orthogonal complement of W.

Sol:

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \sim R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{(reduced row-echelon form)}$$

$\mathbf{w}_1 \quad \mathbf{w}_2$

(a) $W = CS(A)$

$\Rightarrow \quad \{(1,2,1,0),(0,0,0,1)\}$ is a basis for W

(b) $W^{\perp} = (CS(A))^{\perp} = NS(A^{\mathrm{T}})$

$$\because A^{\mathrm{T}} = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2s - t \\ s \\ t \\ 0 \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$\Rightarrow \{(-2,1,0,0) \quad (-1,0,1,0)\}$ is a basis for $W^{\perp}$

- Notes:

(1) $\dim(W) + \dim(W^{\perp}) = \dim(R^4)$

(2) $W \oplus W^{\perp} = R^4$

- Least squares problem:

$$\underset{m\times n}{A}\underset{n\times 1}{\mathbf{x}}=\underset{m\times 1}{\mathbf{b}}$$ (A system of linear equations)

(1) When the system is consistent, we can use the Gaussian elimination with back-substitution to solve for **x**

(2) When the system is inconsistent, how to find the "best possible" solution of the system. That is, the value of **x** for which the difference between $A\mathbf{x}$ and **b** is small.

- Least squares solution:

Given a system $A\mathbf{x} = \mathbf{b}$ of m linear equations in n unknowns, the least squares problem is to find a vector $\hat{\boldsymbol{x}}$ in R^n that minimizes $\|A\mathbf{x}-\mathbf{b}\|$ with respect to the Euclidean inner product on R^m.

Such a vector $\hat{\boldsymbol{x}}$ is called a *least squares solution* of $A\mathbf{x} = \mathbf{b}$.

- Recall:

$A \in M_{m\times n}$

$\boldsymbol{x} \in R^n$

$A\boldsymbol{x} \in CS(A)$ $(CS(A)$ is a subspace of $R^m)$

$\because \mathrm{b} \notin CS(A)$ $(A\boldsymbol{x} = \mathrm{b}$ is an inconsistent system)

Need to find a vector $\hat{\boldsymbol{x}}$ in R^n such that $A\hat{\boldsymbol{x}}$ is closest to **b**.

Let $A\hat{\boldsymbol{x}} = proj_{CS(A)}\boldsymbol{b}$

$\Rightarrow (\boldsymbol{b} - A\hat{\boldsymbol{x}}) \perp CS(A)$

$\Rightarrow \boldsymbol{b} - A\hat{\boldsymbol{x}} \in (CS(A))^{\perp} = NS(A^{\mathrm{T}})$

$\Rightarrow A^{\mathrm{T}}(\boldsymbol{b} - A\hat{\boldsymbol{x}}) = 0$

i.e. $A^{\mathrm{T}}A\hat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$ (the normal system associated with $A\boldsymbol{x} = \mathbf{b}$)

- Summary: ($A\mathbf{x} = \mathbf{b}$ is an inconsistent system)

  The problem of finding the least squares solution of $A\mathbf{x} = \mathbf{b}$ is equal to the problem of finding an exact solution of the associated normal system $A^{\mathrm{T}}A\hat{\mathbf{x}} = A^{\mathrm{T}}\mathbf{b}$.

- Ex 7: (Solving the normal equations)

  Find the least squares solution of the following system

$$A\mathbf{x} = \mathbf{b}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$$ (this system is inconsistent)

and find the orthogonal projection of $\mathbf{b}$ on the column space of A.

- Sol:

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$A^T \mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 11 \end{bmatrix}$$

the associated normal system

$$A^T A\hat{\boldsymbol{x}} = A^T \boldsymbol{b}$$

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} 4 \\ 11 \end{bmatrix}$$

the least squares solution of $A\mathbf{x} = \mathbf{b}$

$$\hat{\boldsymbol{x}} = \begin{bmatrix} -\frac{5}{3} \\ \frac{3}{2} \end{bmatrix}$$

the orthogonal projection of $\mathbf{b}$ on the column space of A

$$\text{proj}_{CS(A)} \boldsymbol{b} = A\hat{\boldsymbol{x}} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} \frac{-5}{3} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} \frac{-1}{6} \\ \frac{8}{6} \\ \frac{17}{6} \end{bmatrix}$$